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Hypothesis

- Image generators show zero-shot depth, segmentation and normal-like outputs, but usually cannot follow exact benchmark formats.
- If a lightly tuned generator reaches specialist-level perception without forgetting generation, its base representation is generalist.
- The proposed analogy: image-generation pretraining can play the same role as next-token pretraining for LLMs.

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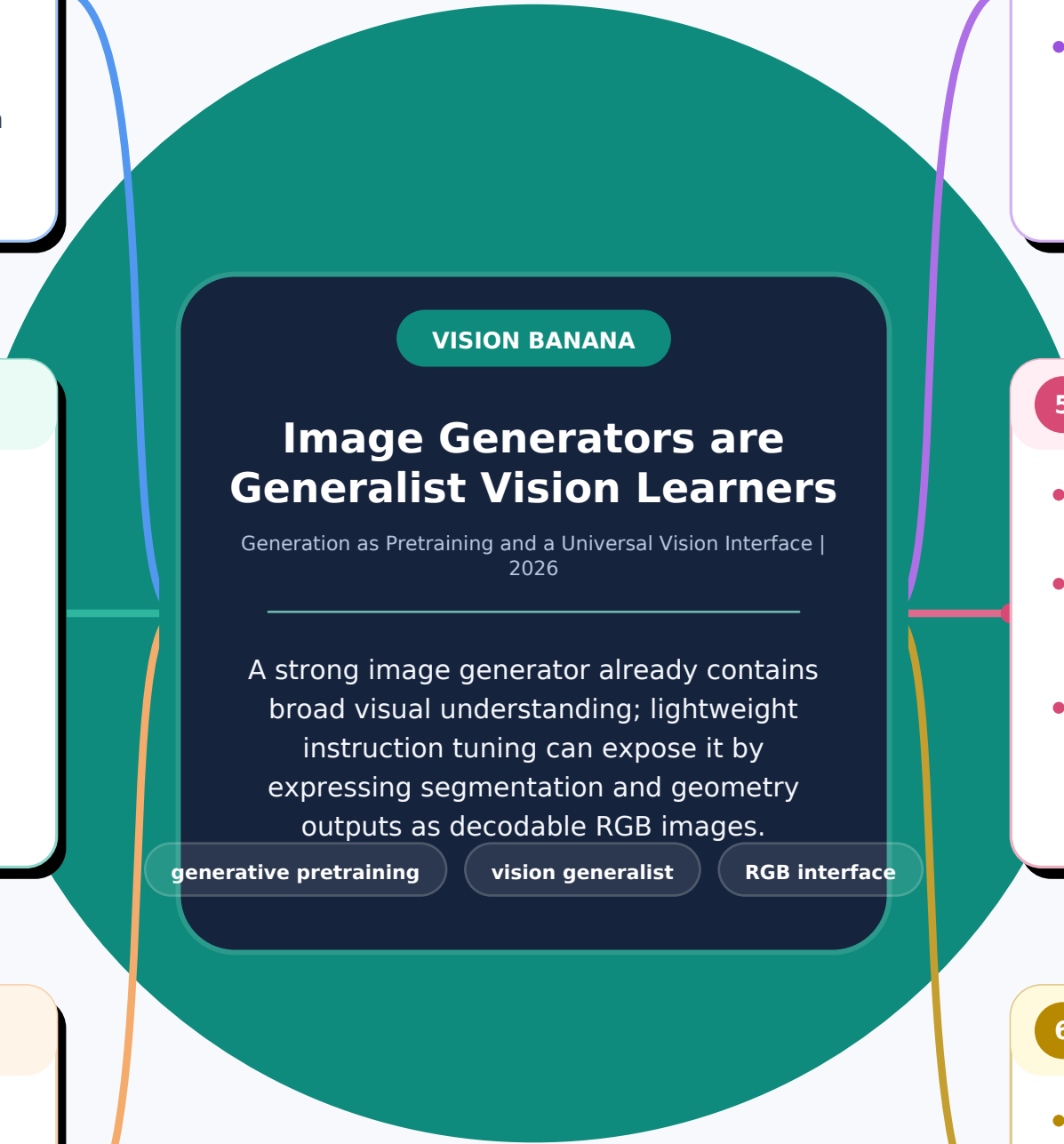
Vision Banana

- Start from Nano Banana Pro and mix a small ratio of vision-task examples into its original generation training mixture.
- A single set of weights supports text-to-image, editing, segmentation, metric depth and surface normals; only the prompt changes.
- Evaluation benchmark training sets are excluded, preserving a strict zero-shot transfer protocol.

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Universal interface

- Prompts request precisely formatted RGB visualizations that can be decoded back into labels, masks or physical geometry.
- Colors encode semantic/instance masks; an invertible curved color map encodes unbounded metric depth.
- Generative outputs naturally represent multiple plausible modes, reducing the need for task-specific heads and losses.



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2D + 3D tasks

ONE MODEL, MANY OUTPUTS

- 2D suite: semantic, instance and referring-expression segmentation, including open-vocabulary noun phrases.
- 3D suite: monocular metric depth and surface-normal estimation using synthetic geometry supervision.
- The model can combine language grounding, semantic grouping and geometric priors through the same image-generation channel.

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Evidence

RIVALS SPECIALISTS

- RefCOCOg cloU 73.8, ReasonSeg gloU 79.3, Cityscapes mIoU 69.9 and SA-Co/Gold cgF1 47.5.
- Metric-depth average delta1 is 0.929 versus Depth Anything 3 at 0.918; surface-normal mean error is 18.928 versus Lotus-2 at 19.642.
- Generation is retained: 53.5% text-to-image and 47.8% editing win rates against the untuned base model.

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Implications & limits

A VISION PARADIGM SHIFT

- Generative vision pretraining may become the foundation for both visual creation and understanding.
- Future work spans more tasks, multiview/video inputs, temporally aware generators and tighter LLM integration.
- Large image generators remain much more computationally expensive than lightweight specialists; deployment needs acceleration.